

Фінальні правки

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Використання індексів у функції WHERE:

Було:

```
if op == "=" and index_name in indexes:
    print(f"Index used for WHERE: {index_name}")

    tree = indexes[index_name]
    row_ids = tree.search(val_clean)

    result = []
    for rid in row_ids:
        if rid in values:
            result.append(values[rid])

    return columns, result
```

Стало:

```
if op == "=" and index_name in indexes:
    print(f"Index used for WHERE: {index_name}")
    tree = indexes[index_name]
    row_ids = tree.search(val_clean)
    result = []
    for i in row_ids:
        val = values[i]
        result.append(val[2])
```

Використання індексів у функції JOIN_ON:

Було:

```
index_name = f"{table_name_2}_{column_2}"
global indexes

if index_name in indexes:
    print(f"Index used for JOIN: {index_name}")

    tree = indexes[index_name]
    joined_rows = []

    for r1 in rows1:
        key = r1[i1]
        matching_ids = tree.search(key)
        for rid in matching_ids:
            if rid in values:
                r2 = values[rid]
                joined_rows.append(r1 + r2)

    joined_cols = cols1 + cols2
    return joined_cols, joined_rows
```

Стало:

```

index_name = f"{table_name_2}_{column_2}"
global indexes
if index_name in indexes:
    print(f"Index used for JOIN: {index_name}")
    tree = indexes[index_name]
    joined_rows = []
    for r1 in rows1:
        key = r1[i1]
        matching_ids = tree.search(key)
        for i in matching_ids:
            val = values[i]
            joined_rows.append(r1 + val[2]);
    return cols1 + cols2, joined_rows

```

Результати тесту з оновленим кодом:

```

NO INDEXED
Running file without arrey...
Running file with arrey...
SELECT_ALL_JOIN: no_arrey=0.000536s    arrey=0.000529s
SELECT_JOIN_ON: no_arrey=0.000536s    arrey=0.000536s
SELECT_WHERE_<: no_arrey=0.000536s    arrey=0.000529s
SELECT_WHERE_=: no_arrey=0.000536s    arrey=0.000524s
SELECT_WHERE_JOIN: no_arrey=0.000525s    arrey=0.000512s
FASTER VERSION:
with arrey
INDEXED
Running file without arrey...
Running file with arrey...
SELECT_ALL_JOIN: no_arrey_i=0.000659s    arrey_i=0.000517s
SELECT_JOIN_ON: no_arrey_i=0.000540s    arrey_i=0.000512s
SELECT_WHERE_<: no_arrey_i=0.000537s    arrey_i=0.000511s
SELECT_WHERE_=: no_arrey_i=0.000524s    arrey_i=0.000509s
SELECT_WHERE_JOIN: no_arrey_i=0.000521s    arrey_i=0.000508s
FASTER VERSION:
with arrey

```